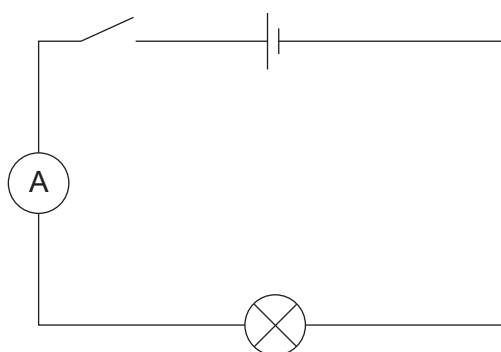


Answer **all** questions.

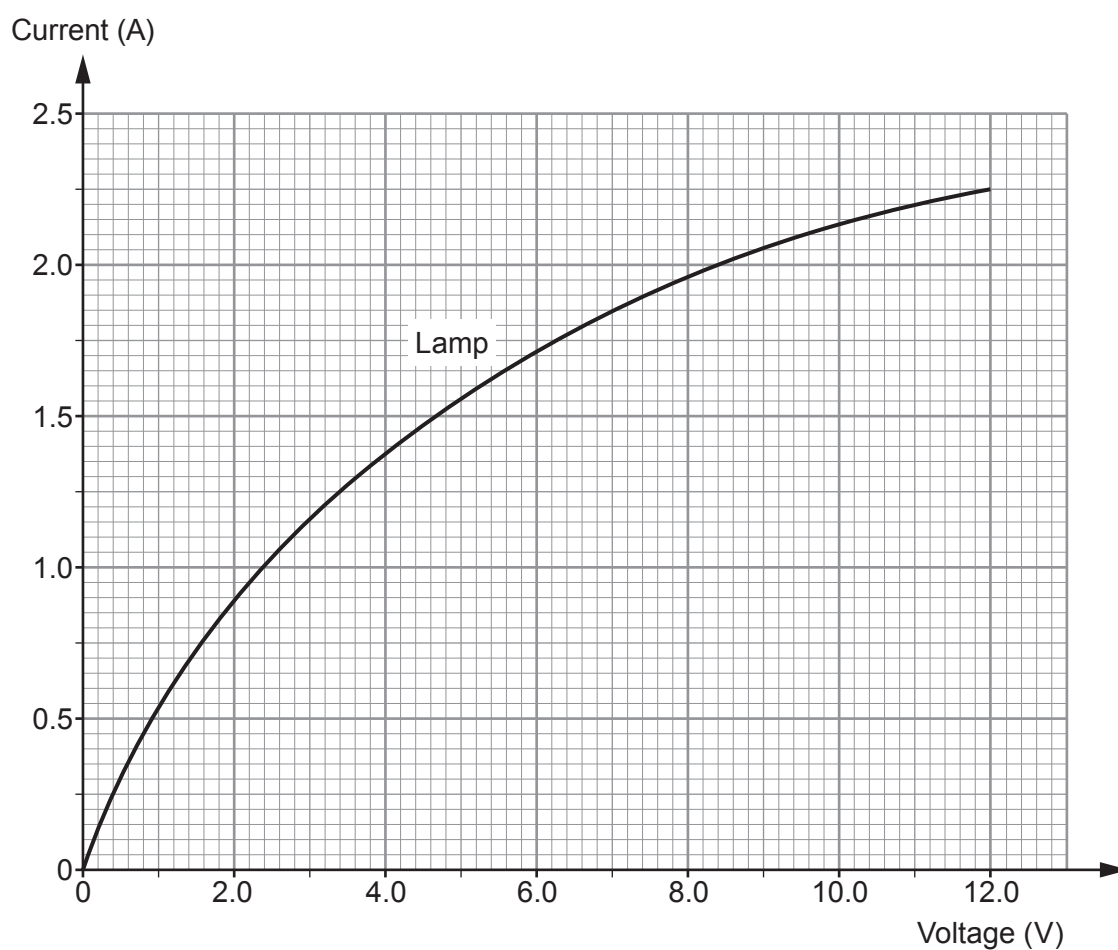
1. A group of pupils investigate the current-voltage ( $I$ - $V$ ) characteristics of different components.

- (a) The first component they investigate is a lamp.  
Part of the circuit used is shown below.  
**Add** a variable resistor **and** voltmeter to the circuit diagram.

[2]



- (b) They draw a graph from their results for the **lamp**. It is shown below.



- (i) One student suggests that as the current through the lamp **doubles** the voltage **triples**.  
Use pairs of data within the range 0.5 A to 2.0 A from the graph to explain whether you agree with the student. [3]
- .....
- .....

- (ii) Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

and information from the graph to calculate the **maximum** power produced by the lamp. [3]

Power = ..... W

- (c) The experiment is repeated with a  $6\ \Omega$  resistor but the results are lost.

- (i) Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

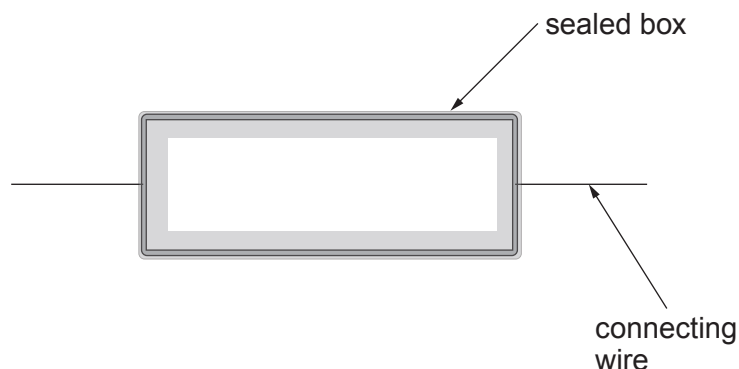
to calculate the current through the  $6\ \Omega$  resistor at 12 V. [2]

Current = ..... A

- (ii) **Draw the line for this resistor** on the grid on the previous page. [1]



- (d) The students are given a sealed box containing another component.



They are asked to confirm whether the hidden component is a diode.

- (i) Describe how they would use the circuit shown in part (a) to confirm whether or not it is a diode. [3]

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- (ii) **Sketch** the  $I$ - $V$  graph that you would expect to obtain for a diode on the grid below. [1]

